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SUPERSEDING
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FEDERAL SPECIFICATION

ADHESIVE, NATURAL OR SYNTHETIC NATURAL RUBBER

This Specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies

1 SCOPE AND CLASSIFICATION

1 1 Scope This specification covers two part natural or synthetic natural rubber adhesives, primarily for the manufacture and repair of articles made of materials coated with natural or synthetic-natural rubber

1 2 Classification Adhesives shall be of the following classes as specified (see 6 2)

1 2 1 Classes

1 - Heat cure for manufacture
2 - Room temperature cure for manufacture
3 - Room temperature cure for repair

2 APPLICABLE DOCUMENTS

2 1 Government publications The issue of the following documents, in effect on date of invitation for bids or solicitations for offers, form a part of this specification to the extent specified herein

AMSC N/A FSC 8040

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FEDERAL SPECIFICATIONS

TT-T-548	Toluene, Technical
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-665	Boxes, Paperboard, Metal Edged and Components
PPP-B-676	Boxes, Setup
PPP-C-96	Cans, Metal, 28 Gage and Lighter

FEDERAL STANDARDS

Fed Std No 123	Marking for Domestic Shipment (Civil Agencies)
Fed Std No 313	Material Safety Data Sheets, Preparation and Submission of

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, Handbooks and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U S Government Printing Office, Washington, DC 20402)

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MILITARY SPECIFICATIONS

MIL-C-23070	Cloth, Laminated and Tape, Coated Cloth, Polyisoprene, Natural or Synthetic Rubber on Nylon

MILITARY STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

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(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

FEDERAL REGULATIONS

Department of Transportation
Code of Federal Regulations
49 CFR-Transportation

(Application for copies of the Department of Transportation Code of Federal Regulations (CFR) should be addressed to the Superintendent of Documents, U S Government Printing Office, Washington, DC 20402)

3 REQUIREMENTS

3 1 Qualification The adhesive furnished under this specification shall be a product which has been tested, and has passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable Qualified Products List (See 4 3 and 6 3)

3 2 Materials The ingredients used in the manufacture of the adhesive shall be of high quality, suitable for the purpose, and shall not cause deterioration of the adherends The manufacturer of the adhesive shall certify to the Government that only natural or synthetic-natural rubber has been used

3 2 1 Toxicity The cleaning compound shall have no adverse effect on the health of personnel, when used for its intended purpose Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as adviser to the contracting agency The manufacturer at time of qualification shall certify that the cleaning compound contains no substance known to be toxic to the user under normal conditions of use Material safety data sheets shall be prepared and submitted in accordance with FED-STD-313, one copy of which shall be forwarded to the preparing activity of the specification and one copy to the qualifying laboratory

3 2 2 Formulation changes The adhesive shall be approved only for the formulation for which qualification tests have been made Subsequent changes to material such as adding pigments, hardeners, solvents, fillers, or any other changes in formulation will require separate qualification testing The manufacturer may submit modified adhesive for approval under this specification using a new manufacturer's designation

3.3 Form The adhesive furnished under this specification shall be furnished as a two-part system consisting of the base adhesive and accelerator Each component shall be packaged separately, in proportioned amounts to yield the optimum physical and mechanical properties when the entire contents of the containers are mixed

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3 4 Physical properties The physical properties of the adhesives shall conform to Table I

3 4 1 Brushability Classes 1 and 2 adhesives shall be readily brushable (i e , flow easily and cover surfaces evenly) when mixed and applied in accordance with the manufacturer's instructions and 4 6 1 Class 3 adhesive shall be readily brushable without thinning in the "as received," "after accelerated storage," and "during working life" (6 hours after mixing) conditions

3 4 2 Working life Classes 1, 2 and 3 adhesives shall have a minimum working life of 6 hours, when prepared and tested as specified in 4 6 2

3 4 3 Accelerated storage Class 3 adhesives, when stored as specified in 4 6 5, shall not gel or deteriorate and shall be brushable

TABLE I Physical and mechanical properties

Properties	Requirements	
	Classes 1 and 2	Class 3 1/
Total solids (percent)	As determined 2/	As determined 2/
Viscosity as received (centipoises)	As determined 2/	As determined 2/
After accelerated storage (centipoises)	--	3/
Seam strength, min		
As received kg/5cm (lbs /2-in width)	195 (425)	195 (425)
After bond aging (percent of as received value, min)	36 3 (80)	36 3 (80)
Strip adhesion		
As received kg/2 5cm (lbs /lin) width)	2 4 (5)	2 4 (5)
After bond aging (percent of as received value, min)	36 3 (80)	36 3 (80)
After accelerated storage (kg/2 5 cm lbs /in width min)	-	2 4 (5)
After water immersion (kg/2.5 cm lbs./in width min)	2 4 (5)	2 4 (5)
Dead load 60C, (140F) 24 hours, 34 kg (75 lbs)	No separation, no creep	No separation, no creep

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1/ Adhesives qualifying for Class 3 will automatically qualify for Class 2.

2/ "As determined" denotes that the value shall be determined during qualification testing The quality conformance test values shall not deviate from the original "as determined" values by more than the amount specified in Note 1 of Table II

3/ Shall not vary from "as received" viscosity by more than ± 20 percent

3 4.4 Shelf storage life (Class 3) The Class 3 adhesives as furnished by the manufacturer shall show no evidence of gelling, hardening, or stringing and shall be capable of meeting the minimum "as received" seam strength requirements of Table I after storage for one year from date of manufacture at room temperature (see 4 5 1 and 4 6 9)

3 5 Workmanship The adhesives shall be homogeneous, free of foreign matter and shall be prepared in accordance with the best commercial practices for this material and shall meet all the requirements of this specification

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection The contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4 2 Classification of inspection The inspection requirements specified herein are classified as follows

a Qualification inspection (see 4 3)
b Quality conformance inspection (see 4 4)

4.3 Qualification inspection Qualification inspection shall consist of all the tests and examinations in Table II

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TABLE II Qualification Inspection

Characteristic	Requirement	Test
Total solids	Table I	4 6 4
Viscosity	Table I	4 6 3
Seam Strength	Table I	4 6 6
Strip Adhesion	Table I	4 6.7
Dead Load	Table I	4.6 8
Brushability	3 4 1	4 6 1
Working Life	3 4 2	4 6.2
Accelerated Storage (Class 3)	3 4 3	4 6 5
Shelf Life (Class 3)	3 4 4	4 6 9

4.3 1 Qualification sample Qualification test samples shall consist of 4-liters (4 quarts) of each class of adhesive, together with the required accelerator Also a sufficient amount of natural rubber coated fabric conforming to variety D of MIL-C-23070, or equal, for use in tests, shall accompany the qualification samples The samples shall be forwarded to the activity designated in the authorizing letter The adhesives shall be furnished in containers of the type specified in Section 5 They shall be plainly and durably marked with the following information

Sampled for qualification tests
Specification MMM-A-139B
Class
Date of manufacture
Name and address of the manufacturer
Manufacturer's identification
Submitted by (name) (date) for qualification tests in accordance with MMM-A-139B under authorization (reference authorizing letter)

4 3 2 1 Manufacturer's data Two copies of the manufacturer's test report, containing complete test data showing that material submitted for qualification conforms to this specification, shall be submitted with the qualification test samples Location and identify of the plant which produced the sample shall also be supplied The contractor shall furnish toxicological data (see 3 2 1) necessary to evaluate the adhesive for proposed use along with the certification required in 3 2

4 3 3 Retention of qualification In order to retain qualification of an adhesive approved for listing on the Qualified Products List (QPL), the manufacturer shall verify by certification to the Qualifying Activity that the manufacturer's product complies with the requirements of this specification The time of periodic verification by certification shall be in two year intervals and shall be initiated by the Government The Government reserves the right to re-examine the qualified product to determine that the product continues to meet any or all specification requirements

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4.4 Quality conformance

4.4.1 Lot formation. A lot shall consist of a batch of adhesive of one class manufactured from the same components processed at the same time No batch of adhesive will be considered for acceptance under a contract or order, if more than 30 days have elapsed from the date on which the actual manufacturing process in completed to the date on which the batch is made available to the Government for selection of samples

4.4.1.1 Source inspection Unless otherwise specified, all materials (each lot) furnished to this specification shall be source inspected before shipment to the Government

4.4.2 Sampling and Inspection

4.4.2.1 Physical properties Two containers shall be selected at random from each lot Material from each container shall be subjected to the testing in Table III If a test specimen fails to meet any of the tests specified by this specification, the lot represented by the sample shall be rejected

TABLE III Quality conformance physical properties

Characteristic	Test Paragraph	Requirement Paragraph
Total solids 1/	4.6.4	Table I
Viscosity:	4.6.3	Table I
As received 1/		
After accelerated storage		
Working life 1/	4.6.2	3.4.2
Brushability	4.6.1	3.4.1
As received		
After accelerated storage		
Seam strength (as received)	4.6.6	Table I
Strip adhesion (as received)	4.6.7	Table I
Dead load	4.6.8	Table I

1/ The test values for all classes shall not deviate from the original "as determined" values by more than ± 10 percent for total solids, ± 20 percent for viscosity (as received) Example for applying tolerance on total solids The value of an original total solids of 8 percent is from 7.2 to 8.8 percent

4.4.2.2 Packaging inspection

4.4.2.2.1 Visual inspection of filled containers A random sample shall be selected from each batch in accordance with MIL-STD-105 at Inspection Level I and an Acceptable Quality Level (AQL) of 2.5 percent defective to verify conformance to all requirements regarding fill, components (accelerator), closure, marking, and other requirements not involving tests

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4 4 2 2 2 Examination for packaging and marking An examination shall be made to determine that packaging and marking comply with the requirements of Section 5 of this specification Defects shall be scored in accordance with the list below The sample unit for this examination shall be one shipping container fully prepared for delivery except that it shall not be palletized and need not be sealed Shipping containers fully prepared for delivery that have not been palletized shall be examined for defects of closure The lot size shall be the number of shipping containers in the end item inspection lot The samples for this examination shall be selected at random in accordance with MIL-STD-105, inspection level S-2 and an acceptable quality level (AQL) of 4 0 defects per hundred units

EXAMINE	DEFECT
Packaging	Container not as specified, closures not accomplished by specified or required methods or materials Leakage or seepage of contents Non-conforming component, component missing, damaged or otherwise defective Bulged or distorted container
Markings	Data, including directions for use, omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements

4 4 2 2 3 Examination for palletization An examination shall be made to determine that palletization complies with the requirements of Section 5 of this specification Defects shall be scored in accordance with the list below THe sample unit shall be one palletized unit load fully prepared for delivery The lot size shall be the number of palletized unit loads in the end item inspection lot The samples for this examination shall be selected at random in accordance with MIL-STD-105, inspection level S-1 and an acceptable quality level (AQL) of 6 5 defects per hundred units

EXAMINE	DEFECT
Finished dimension	Length, width, or height exceeds specified maximum requirements
Palletization	Not as specified Pallet pattern not as specified Interlocking of loads not as specified Load not banded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted, incorrect, illegible, of improper size, location, sequence or method of application

4 5 Test conditions

4 5 1 Standard temperature of tests Unless otherwise specified, tests shall be conducted at a temperature of 24°±3°C (75°±5°F) and 50 ± 5 percent relative humidity

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4 5.2 Application instructions for all testing. The following procedures shall be performed:

a Using a clean lint-free cloth each time, clean all surfaces to be bonded by washing three times with a cloth saturated with toluene conforming to TT-T-548 Allow to dry for a minimum of 15 minutes before applying the first coat of adhesive

b Stir material (base and accelerator) well before and after combining as recommended by the manufacturer Class 3 adhesive shall not be thinned.

c. For classes 1 and 2, the number of coats (not to exceed three to each surface), the application of the adhesive, the drying time between coats and drying time before bonding shall be as specified by the manufacturer. For Class 3, at least two and not more than three coats of adhesive shall be applied to each surface to be bonded Allow 15 minutes drying time between each coat Before bonding, allow the final coat to dry 15 minutes or until the film develops an aggressive tack but will no longer transfer to knuckle

d Roll the bonded seams together by applying as much manual force as possible with approximately a 1 Kilogram (2-pound) roller having a diameter of approximately 5cm (2 inches) and a width of approximately 5cm (2 inches)

e Class 3 seam and strip specimens shall be cured for 5 days at 24°±3°C (75°±5°F), and 50±5 percent relative humidity before testing Classes 1 and 2 seam and strip specimens shall be cured according to manufacturer's instructions before testing The manufacturer's recommended curing time shall not exceed 5 days If elevated temperature curing is recommended by the manufacturer, the specimens shall be conditioned for at least 16 hours at 24°±3°C (75°±5°F), and 50±5 percent relative humidity prior to testing

4 6 Test procedures

4 6 1 Brushability Classes 1, 2 and 3 adhesive supplied by the manufacturer and mixed, when required, shall be brushed to determine conformance with 3 4 1

4 6 2 Working life The two-part adhesive shall be prepared by mixing the base and accelerator in accordance with the manufacturer's instructions, and allowed to stand at 24°±3°C (75°±5°F) and 50±5 percent relative humidity The adhesive shall be stirred at 30 minute intervals and brushability determined immediately after stirring A significant decrease in brushability shall be construed as the expiration of working life (see 3 4 2)

4 6 3 Viscosity

4 6 3 1 Apparatus The apparatus used shall be Brookfield viscosimeter, Model RVF, or equivalent

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4 6.3 2 Procedure. Viscosity determinations shall be made using a 1-quart sample of adhesive Inspection results shall be obtained directly from the original quart containers used in filling of the contract or purchase order The adhesive shall be at a uniform temperature of 24°±3°C (75° ±5F) during the test The adhesive shall be thoroughly stirred to uniform consistency immediately before testing Reading shall be taken when the pointer assumes a steady position after release of the clutch (see 3 4)

4 6 4 Total solids

4 6 4 1 Apparatus An assembly consisting of an aluminum foil dish approximately 6.4cm (2 5 inches) in diameter and 2.0cm (0 75 inches) in depth, and quick-closing airtight container and lid of slightly greater dimension than the aluminum foil dish shall be used for this test

4 6 4 2 Procedure The assembly (dish, container, and lid) shall be weighed After weighing, the lid shall be removed and approximately 3 grams of thoroughly stirred adhesive shall be placed in the aluminum dish as soon as possible after stirring The lid shall be immediately replaced and the assembly reweighed The aluminum dish containing the adhesive shall be removed from the container and dried in a circulating air oven for 24±1 hours at 70°±1°C (158°±2°F) After the drying period, the aluminum dish shall be cooled in a desiccator for 1 hour, the lid replaced and the assembly reweighed The total solids shall be calculated as follows

$$\text{Percent total solids} = \frac{\text{Net dry weight of sample}}{\text{Net wet weight of sample}} \times 100$$

All weighing shall be made to the nearest milligram Total solids shall be reported as the average of three determinations (see 3 4)

4 6 5 Accelerated storage A 1-quart sample of Class 3 adhesive and proportionate quantity of the accompanying accelerator, when required, in the containers used in filling contracts or orders, shall be placed unopened in a circulating air oven for 15 days at 60°±1°C (140°±2°F) At the end of the storage period, the adhesive and accelerator shall be removed from the oven and cooled overnight at 24°±3°C (75°±5°F) The containers shall then be opened and the materials examined for gelling or other evidence of deterioration and tested for conformance with the brushability and viscosity requirements (see 3 4 1, Table I and 4 6 7 2 4)

4 6 6 Seam strength

4 6 6 1 Fabrication of laminates Twelve pieces, each 30 by 13 cm (12 by 5 inches), of natural rubber-coated fabric conforming to variety D of MIL-C-23070, or equivalent, shall be bonded to make up 6 laminates Cleaning, drying time between coats and before bonding, and bonding of fabric shall be as specified in 4 5 2 A minimum of 16 hours shall elapse before using laminates to prepare test seams The rubber coated fabric shall be furnished by the manufacturer

4 6 6 2 Preparation of test seams Seam specimens shall be prepared for "as received", "bond aging", and "dead load" tests by forming 2cm (0 75 inch) lap seams along the 30cm (12-inch) dimension Cleaning, drying time between coats and before bonding, bonding of fabric, and curing and conditioning time

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for seams shall be as specified in 4 5.2. Upon completion of the conditioning period, sufficient 5cm (2-inch) wide seam test specimens shall be cut from the coated fabric, discarding 2 5cm (1-inch) strips from each end of the fabric

4 6 6 3 Apparatus The apparatus shall consist of a power-driven inclined plane, pendulum, strain gage, or torsion bar type testing machine The capacity of the testing machine shall be such that the tension at failure shall not be more than 85 percent nor less than 15 percent of the full scale load The rate of travel of the power-actuated jaws shall be 5cm (2 inches) per minute The distance between the clamps shall be 7 5cm (3 inches) at the start of the test

4 6 6 4 Procedure "As received" seam strength tests shall be conducted on seams cured as specified in 4 5 2 Bond-aged seam strength tests shall be conducted, after proper curing time has elapsed, as follows The seams shall be aged for 7 days at 70°±1°C (158°±2°F), and then allowed to remain at room temperature for 24 hours prior to test Three specimens of each test condition shall be tested and results averaged.

4 6 7 Strip adhesion

4.6 7 1 Preparation of test specimens Seams for strip adhesion shall be prepared by bonding 10cm by 15cm (4-by-6 inch) pieces of fabric over the whole area, except for a 2 5cm (1 inch) margin along the entire 10cm (4 inches) of one end Cleaning, drying time between coats and before bonding, bonding of fabric, curing and conditioning time shall be as specified in 4 5 2 After conditioning, three 2 5cm (1 inch) wide specimens shall be cut from the bonded fabric, discarding 1cm (0 5 inch) strips from each side of the fabric Three specimens shall be required for each of the following tests as received, bond aging, water immersion, and after accelerated storage

4.6 7 2 Apparatus The apparatus shall consist of a power-driven inclined plane, pendulum, strain gage, or torsion bar type testing machine The capacity of the testing machine shall be such that the tension at failure shall not be more than 85 percent more or less than 15 percent of the full scale load The machine shall be autographic In the pendulum type machine, the weight level shall swing as a free pendulum without engagement of pawls

4 6 7.2 1 As received The 2 5cm (1 inch) wide strip adhesion specimens shall be held in the testing machine by grips which clamp firmly and prevent slippage at all times The rate of travel of the power actuated grip shall be 5cm (2 inches) per minute The specimen shall be stripped at an angle of approximately 90° and the separation continued for a distance of at least 7 5cm (3 inches) Result shall be recorded as the average of the three specimens tested

4 6 7 2 2 Bond aging Bond aging specimens, prepared as specified in 4 6 7 1, shall be aged for 7 days at 70°±1°C (158°±2°F.) and 50±5 percent relative humidity and then allowed to remain at room temperature for 24 hours prior to testing as specified in 4.6 7.2 1.

4 6 7 2 3 Water immersion. Water immersion specimens prepared as specified in 4 6 7 1 shall be immersed in distilled water for 48 hours at room temperature with the 1 inch nonadhered portion of the specimen above the water

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level The specimens shall be tested as specimens in 4 6 7 2 1 within 5 minutes after removal from the water

4 6 7 2 4 After accelerated storage Accelerated storage specimens shall be prepared as specified in 4 6 7 1 from adhesives which have been subjected to the accelerated aging test of 4 6 5 The specimens shall be cured as specified for Class 3 adhesives (4 5 2(e)) before testing for compliance with the requirements of this specification

4 6 8 Dead load Seam strength specimens as prepared in 4 6 6 2 shall be used One end of the specimen shall be clamped firmly to a stationary member that will support the required load in a circulating air oven maintained at 60°±1°C (140°±2°F) A clamp shall be attached to the other end of the specimen and a load of 75 pounds suspended from the clamp The load shall be supported by the specimen without evidence of seam separation or creep for 24 hours Triplicate tests shall be conducted

4 6 9 Shelf storage life (class 3) One quart package of adhesive, properly packaged (see 5 1), being submitted for qualification for Class 3 under this specification shall be stored under conditions specified in 3 4 5 At the end of the storage period, the adhesives shall be visually examined for any obvious evidence of deterioration The samples shall then be used in the preparation of strip adhesion specimens (4 6 7) and tested in accordance with 4 6 7 2 1 for compliance with the requirements of 3 4 5

5 PACKAGING

5 1 Preservation Preservation shall be Level A, B or Commercial, as specified (see 6 2 1)

5 1.1 Level A

5 1 1 1 Base adhesive Base adhesive shall be packaged in round multiple friction top cans conforming to Type V, Class 2 of PPP-C-96 Cans shall be standard U S pint, quart or gallon

5 1 1 2 Accelerator Accelerators shall be packaged in round multiple friction top cans conforming to Type V, Class 2 of PPP-C-96, or in suitable dark bottles having metal screw top caps with an inner metal foil backed gasket, The caps shall be supplemented by cellulose caps or band to reduce evaporation

5 1 1 3 Kit Material packaged as specified in 5 1 1 1 and 5 1 1 2 shall be packaged as a kit in boxes conforming to W6C or W6S of PPP-B-636 or variety 4, style, type and class optional of PPP-B-566 Each box shall be waterproofed as specified for the waterproofing of slotted boxes in accordance with the appendix of PPP-B-636

5 1 1 4 Intermediate package Kits packaged as specified in 5 1 1 3 shall be intermediate packaged in quantities of 4, 6 or 12 kits in boxes conforming to requirements of PPP-B-636, V3c

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5 1 2 Level B. (Civil Agencies). Packaging for Level B shall be as in 5 1 1 except that boxes conforming to PPP-B-665 or PPP-B-676 may also be used, and the boxes conforming to PPP-B-636 shall be class domestic (see 6 2).

5 1 3 Level C The adhesive shall be packaged to afford protection against deterioration and damage during shipment under all environmental conditions When two adhesive components are necessary as a unit, the components shall be packaged as a kit Containers shall conform to the requirements of CFR Title 49

5 2 Packing Packing shall be Level A, B or Commercial as specified (see 6.2)

5 2 1 Level A Adhesives, of one class only, packaged as specified in 5 1 1 2 shall be packed in wood cleated boxes conforming to requirements of PPP-B-601, overseas type, style optional in quantities not to exceed the weight limitation of the container Box closure and strapping shall be in accordance with the appendix to the box specification Strapping shall be zinc coated

5 2 2 Level B.

5 2 2 1 Military requirements Adhesives, intermediate packaged as specified in 5 1 1 3, shall not require any additional packing Boxes shall be strapped in accordance with the requirements in the applicable specification and the appendix thereto. Weight limitations of the respective container specification shall not be exceeded

5 2 2 2 Civil agencies The intermediate packages, 48 one pint, 12 one quart or 4 one gallon cans shall be packed in a close-fitting box conforming to PPP-B-636, Type CF or SF, class domestic, style RSC Closure shall be by Method II of the appendix to PPP-B-636

5 2 3 Level C Adhesive, packaged as in 5 1 shall be packed to assure carrier acceptance and safe delivery to destination at lowest rating in conformance to the rules and regulations of the mode of transportation utilized

5 3 Marking

5 3 1 Civil agencies In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed Std No 123 and shall include the labeling specified in 5 3 2 1 and 5 3 2 2, except that the exterior container shall not contain "Instructions for Use"

5 3 2 Military requirements In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with MIL-STD-129 and shall include the labeling specified in 5 3 2 1 and 5 3 2 2, except that the exterior container shall not contain "Instructions for Use".

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5.3.2.1 Base adhesive Each base adhesive container shall be durably and legibly labeled to include the following information:

ADHESIVE, NATURAL OR SYNTHETIC-NATURAL RUBBER
Specification MMM-A-139B
Class
Manufacturer's name and address
Manufacturer's designation
Batch Number
Volume contained (if base, mixing ratio by volume)
Contract No (order number, if contract number is not assigned)
Manufacturer's batch numbers and designation of required accelerator
Date of manufacture, by month and year (not by code)
Expiration date
Date of shipment
Instructions for use (shall include method of mixing, working time, recommended thinner, drying time between coats, number of coats, drying time before bonding and curing time and temperature)

Precautions to be used
Avoid breathing fumes
Use in ventilated area
Keep away from flame
Replace top tightly

Store in a cool, dark place

5.3.2.2 Accelerator Each accelerator container shall be legibly and durably marked with the following information

ADHESIVE, NATURAL OR SYNTHETIC-NATURAL RUBBER
Specification MMM-A-139B
Manufacturer's name and address
Manufacturer's batch number
Date of manufacture, by month and year (not by code)

Precautions to be used
Volume contained
Accelerator for (insert product designation for base compounds and mixing ratio by volume of accelerator and base)

6 NOTES

6.1 Intended use The natural rubber adhesives are intended for adhering natural or synthetic natural rubber and in particular for the manufacture and repair of articles fabricated from natural rubber coated fabrics These adhesives shall not be used for bonding nylon

6.1.1 Classes 1 and 2 Classes 1 and 2 adhesives must be applied in accordance with the manufacturer's instructions Application procedures used

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to determine conformance of approved Classes 1 and 2 adhesives are available to equipment manufacturers and Government agencies from the Naval Air Development Center (see 6 4)

6 1 2 Class 3 Class 3 adhesive shall be used only for bonding of natural rubber items when used for the repair of life saving equipment

6 2 Ordering data

6 2 1 Acquisition requirements Acquisition documents should specify the following

a Title, number and date of this specification

b. Class of adhesive and capacity of containers (see 1 2 and 5 1 1.1)

c Levels of packaging and packing (see 5 1 and 5 2) Class 3 adhesive should not be purchased in quantities greater than necessary for a 6 months' supply

6 2 1 For civil agency procurements. When Level B packaging is required for civil agency procurements, the Level A requirements in 5 1 2 shall apply

6 3 Storage Elevated storage temperatures shorten the usable life of these adhesives, therefore, adhesives should be stored in a cool place

6 4 Qualification With respect to products requiring qualification awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification The activity responsible for the Qualified Products List is the Naval Air Systems Command, Washington, D C 20361, however, information pertaining to qualification of products may be obtained from the Naval Air Engineering Center, Lakehurst, New Jersey 08733-5100, Attention Systems Engineering and Standardization Department (Code 9321) Class 3 adhesives conforming to the other requirements of this specification may, at the option of the qualifying activity, be placed on the Qualified Products List of products qualified under MMM-A-139 (see 6 4) prior to the completion of the storage life tests Modifications of the manufacturer's storage life recommendations or removal of the adhesive from the Qualified Products List will be necessary, if subsequently the adhesive does not pass the storage life tests

6 5 Subject term (key word) listing.

Adhesive, rubber
Natural rubber
Natural synthetic rubber
Qualification
Rubber to rubber repair
Two-part

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6.6 Asterisks are not used in this revision due to the extensiveness of the changes

MILITARY INTEREST

Custodians

Army - MR
Navy - AS
Air Force - 20

Review activities

Army - AV, GL
Air Force - 84

User activities

Army - ME, AT
Navy - YD, MC, OS, SH

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
DOT - APM
HUD - HCC

Preparing activity

Navy - AS

(Project No 8040-0457)

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